

Cambridge (CIE) IGCSE Chemistry
Paper 1: Multiple Choice (Set C)
Core

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks

/ 40

Instructions

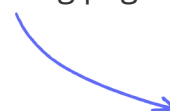
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Materials

- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 The diagram shows a cup of hot chocolate which was prepared by mixing boiling water with chocolate powder.

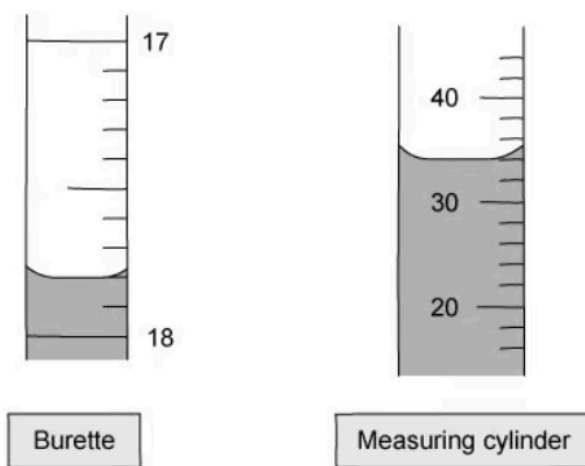


Which row correctly describes the particles of water in the air above the cup compared to those in the hot chocolate drink inside the cup?

	Particles move faster	Particles are closer together
A	✓	✓
B	✓	×
C	×	✓
D	×	×

(1 mark)

- 2 The diagram shows a burette and a measuring cylinder, both of which contain a liquid.



Which row shows the correct reading for both?

	Burette	Measuring cylinder
A	18.4	34
B	18.4	56
C	17.8	34
D	17.8	56

(1 mark)

- 3 Solid P melts at exactly 60°C and boils at exactly 310°C. Solid Q dissolves in ethanol and shows a green colour and a yellow colour when analysed using paper chromatography.

Which row is correct?

	Contains only one substance	Contains more than one substance
A	Q	P
B	Q	-
C	-	Q
D	P	Q

(1 mark)

4 Element X contains 12 of each sub-atomic particle.

What is the correct Group number for element X?

- A.** Group 0
- B.** Group I
- C.** Group VII
- D.** Group II

(1 mark)

5 Copper is sometimes used to make cooking utensils and pans.

Three properties of copper are shown below:

1	Good conductor of electricity.
2	Corrosion resistant.
3	Good conductor of heat.

Which of these properties make it good for making cooking utensils?

- A.** 1 and 3
- B.** 1 only
- C.** 2 and 3
- D.** 2 only

(1 mark)

6 Calcium reacts with fluorine to form calcium fluoride

Which statement describes what happens to calcium atoms in this reaction?

- A.** Calcium ions lose electrons to form positive charges. .
- B.** Calcium atoms gain 3 electrons to form positive ions.
- C.** Calcium ions gain 2 electrons to form positive ions.
- D.** Calcium atoms lose 2 electrons to form positive ions.

(1 mark)

7 The atom of element Z has 6 electrons in its outer shell. How would this element react?

- A.** It shares two electrons with four from another atom to form two covalent bonds
- B.** It shares four electrons with four from another atom to form four covalent bonds
- C.** It shares four electrons with two from another atom to form two covalent bonds
- D.** It shares two electrons with two from another atom to form two covalent bonds

(1 mark)

8 Which statement correctly describes the structure of macromolecules?

- A.** Giant molecular crystal which is held together by weak intermolecular forces.
- B.** Giant molecular crystal which is held together by strong ionic bonds.
- C.** Giant molecular crystal which is held together by weak metallic bonds.
- D.** Giant molecular crystal which is held together by strong covalent bonds.

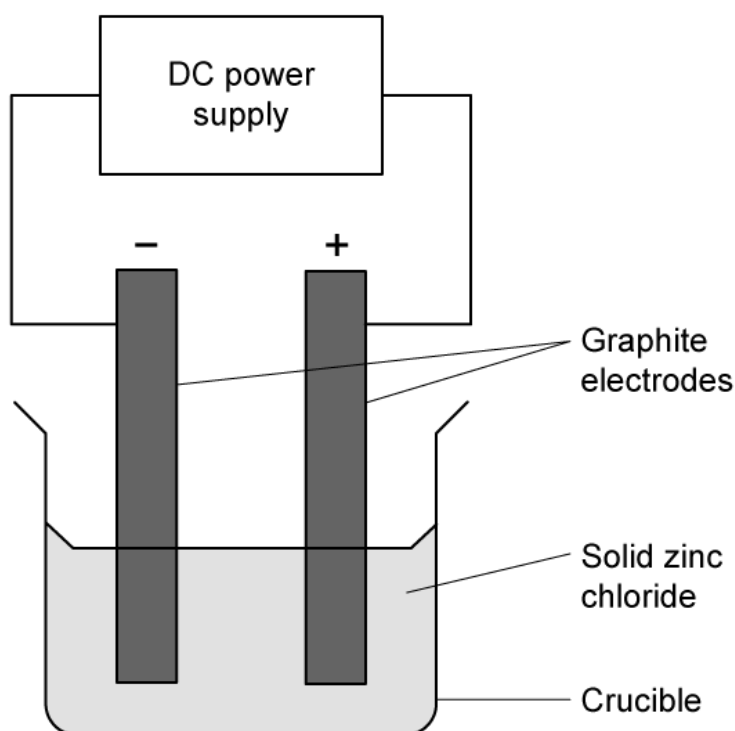
(1 mark)

9 What is the balanced equation for the reaction of methane and oxygen?

- A. $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
- B. $\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$
- C. $2 \text{CH}_4(\text{g}) + 4 \text{O}_2(\text{g}) \rightarrow 2 \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
- D. $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$

(1 mark)

10 A student was investigating the electrolysis of zinc chloride.



What is the correct reason for electrolysis not working?

- A. Solid zinc chloride has been used
- B. The electrodes don't conduct electricity
- C. The power supply is not connected properly
- D. The crucible is too small to allow ions to flow

(1 mark)

- 11** A student adds a small amount of ammonium chloride to a beaker of water. The temperature of the water decreases from 21 °C to 17 °C.

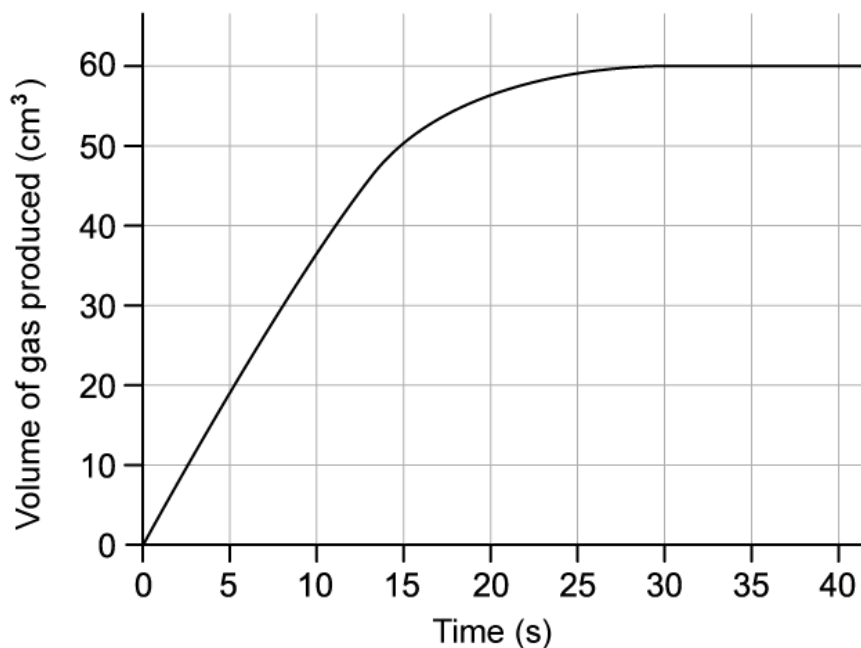
Which type of reaction has occurred and why?

	type of reaction	reason
A	exothermic	heat is released
B	exothermic	heat is absorbed
C	endothermic	heat is released
D	endothermic	heat is absorbed

(1 mark)

- 12** The rate of a reaction was monitored by recording the volume of gas produced every 5

seconds.



What is the mean rate of reaction in the first 10 seconds of the reaction, in cm^3/s ?

- A. 2.9
- B. 3.0
- C. 3.5
- D. 2.5

(1 mark)

- 13 Magnesium metal reacts with an excess of hydrochloric acid solution to form magnesium chloride and hydrogen:



Which of the following will **not** increase the rate of this reaction?

- A. Increase the temperature of the acid
- B. Increase the surface area to volume ratio of the pieces of magnesium
- C. Increase the concentration of the hydrochloric acid
- D. Increase the volume of hydrochloric acid solution used

(1 mark)

14 A student was given an unknown substance and asked to analyse it. The student decided to perform three tests and recorded the results as shown below.

1. On heating with ammonium chloride a gas is released
2. The substance turns methyl orange yellow.
3. The substance does not react with potassium carbonate.

What kind of substance was the student given?

- A.** A salt
- B.** A metal
- C.** A base
- D.** An acid

(1 mark)

15 Samples of two solids, P & Q, are separately dissolved in water and made into solutions.

Which combination of P & Q solutions, would give an insoluble product on mixing?

	P	Q
A	potassium chloride	sodium hydroxide
B	ammonium chloride	potassium sulfate
C	sodium nitrate	magnesium sulfate
D	barium chloride	sodium sulfate

(1 mark)

16 What is the oxidation state of Fe in Fe_2O_3 ?

- A.** +2
- B.** -2
- C.** -3
- D.** +3

(1 mark)

17 The results of two tests on a solid are shown below.

Test	Observation
Acidified silver nitrate added	Cream precipitate formed
Aqueous sodium hydroxide added	Red-brown precipitate formed

What is the solid?

- A.** Iron(II) bromide
- B.** Iron(III) bromide
- C.** Iron(II) chloride
- D.** Iron(III) chloride

(1 mark)

- 18** A student gradually added excess potassium hydroxide from a burette to a conical flask containing sulfuric acid as part of a rate of reaction investigation.

Which statement describes the change in pH of the sulfuric acid as the aqueous potassium hydroxide was added?

- A.** The pH decreases from 14 to neutrality and remains constant
- B.** The pH decreases from 12 to 2
- C.** The pH increases from 1 to neutrality and remains constant
- D.** The pH increases from 1 to around 13

(1 mark)

- 19** Copper(II) sulfate can be prepared by reacting solid copper(II) carbonate with hot sulfuric acid. The copper(II) carbonate is always added in excess.

Why is the solid reactant added in excess?

- A.** To ensure all of the acid reacts.
- B.** To make the reaction go faster.
- C.** To ensure all of the solid reactant has reacted.
- D.** To increase the product yield.

(1 mark)

- 20** Potassium and krypton are elements in Period 4. Potassium is at the beginning of the period and is more metallic than krypton which is at the end of the period.

What makes potassium more metallic than krypton?

- A.** It has less electrons than krypton.
- B.** It has less protons than krypton.
- C.** It has less electron shells.
- D.** It has less outer shell electrons.

(1 mark)

- 21** An element has an electronic configuration of 2,8,6.

Which group and period can the element be found in?

	group	period
A	6	3
B	6	8
C	3	6
D	8	2

(1 mark)

22 Which one of the following names does not correspond to a group of elements on the Periodic table?

- A.** Alkali
- B.** Group VI
- C.** Alloys
- D.** Halogens

(1 mark)

23 Some of the properties of four different elements Q, P, R and S are listed below:

1	Q melts at 1410 °C and forms an acidic oxide.
2	P has a high density and is easily drawn into wires.
3	R acts as a catalyst and its oxide reacts with acids
4	S is a red-brown solid used to make alloys.

Which of the elements are metals?

- A.** 1 and 3 only
- B.** 2 and 4 only
- C.** 2, 3 and 4
- D.** 1,2 and 3

(1 mark)

24 The reactivity of the halogens decreases moving down the group.

Which statement explains this observation?

- A.** Decreasing distance between the nucleus and outer shell makes it easier for the atom to attract electrons.
- B.** Increasing distance between the nucleus and outer shell makes it easier for the atom to attract electrons.
- C.** Decreasing distance between the nucleus and outer shell makes it harder for the atom to attract electrons.
- D.** Increasing distance between the nucleus and outer shell makes it harder for the atom to attract electrons.

(1 mark)

- 25** Calcium, copper, iron and magnesium are metals. They can be placed in order of reactivity.

Which of the following statements about these metals is correct?

- A.** Iron reacts with steam but magnesium does not.
- B.** Copper reacts with dilute hydrochloric acid to form copper (II) chloride.
- C.** Magnesium and calcium both react with hot water.
- D.** Iron (II) oxide cannot be reduced by heating strongly with carbon.

(1 mark)

- 26** Which statement about the extraction of iron from its ore is **not** correct?

- A.** Iron can be extracted by reduction with carbon
- B.** Iron is easier to extract than zinc
- C.** Iron is easier to extract than copper
- D.** Iron is extracted from the ore, hematite

(1 mark)

- 27** The table shows some properties of four metals

Which of the four metals would be the most suitable for aircraft construction?

	density	strength	resistance to corrosion
A	high	low	low
B	low	high	high
C	high	high	low
D	low	low	high

- A.** Option A
- B.** Option B
- C.** Option C
- D.** Option D

(1 mark)

28 Some metals are unreactive and exist on their own in the earth.

What are these metals known as?

- A.** Transition Metals
- B.** Native metals
- C.** Alkali Metals
- D.** Alkaline earth metals

(1 mark)

29 The table below shows the composition of the atmosphere found on a new planet.

gas	percentage by volume
nitrogen	74
carbon dioxide	4
oxygen	22

Which gases are present in the atmosphere of the planet in a lower percentage than they are in the Earth's atmosphere?

- A.** oxygen and carbon dioxide
- B.** nitrogen and oxygen
- C.** nitrogen only

D. carbon dioxide only

(1 mark)

30 Which words complete the gaps 1 & 2 in the word equation for photosynthesis?

carbon dioxide + 1 → 2 + oxygen

	1	2
A	water	hydrogen
B	glucose	water
C	hydrogen	carbon monoxide
D	water	glucose

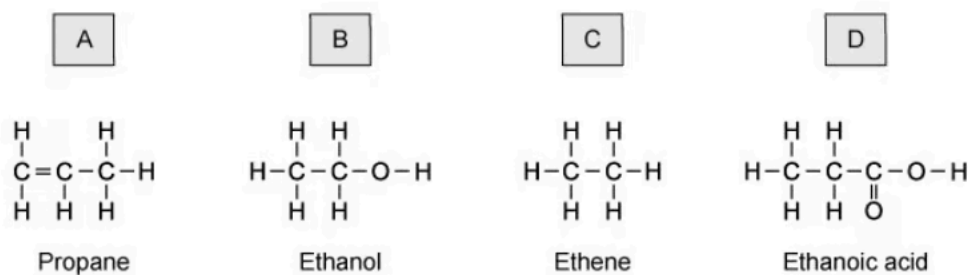
(1 mark)

31 Which strategy for reducing the effect of environmental issues would **not** help reduce climate change?

- A.** Reduce the amount of livestock farming
- B.** Plant more trees
- C.** Fit catalytic convertors on vehicles
- D.** Use renewable energy resources

(1 mark)

32 Which one of the following structures is correctly named?



(1 mark)

33 Which of the following properties of different compounds in petroleum enable it to be separated into fractions?

1	Chain length
2	Chemical reactivity
3	Solubility in water
4	Boiling point

A. 1 & 3

B. 1 & 4

C. 2 & 3

D. 2 & 4

(1 mark)

- 34** A student is provided with solutions of dilute hydrochloric acid, dilute sulfuric acid, solid magnesium oxide, zinc carbonate and copper.

How many different salts could the student make?

- A.** 2
- B.** 3
- C.** 4
- D.** 6

(1 mark)

- 35** Ethanol is used as a solvent and fuel, it is produced by the1..... of2..... .

Which words would correctly complete the gaps 1 and 2?

	1	2
A	combustion	ethane
B	combustion	glucose
C	fermentation	glucose
D	fermentation	ethane

(1 mark)

- 36** Hydrocarbons separated by fractional distillation of petroleum can be cracked to make useful shorter chain products.

Which substance is not a product of the cracking of propane, M_r 44?

- A.** C_4H_8
- B.** C_2H_4
- C.** C_3H_6
- D.** H_2

(1 mark)

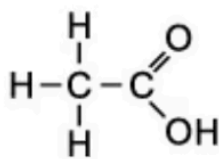
- 37** Butene and octene belong to the same homologous series.

What would be the same for butene and octene?

- A.** Boiling point
- B.** Number of hydrogen atoms per molecule
- C.** Relative molecular mass
- D.** Functional group

(1 mark)

- 38** The diagram shows the structure of a compound, **X**.



Which row correctly describes the properties of **X**?

	colourless	characteristic smell	dissolves in water
A	✓	x	✓
B	✓	✓	x
C	✓	✓	✓
D	x	x	✓

(1 mark)

39 Which of the following elements forms an oxide that can be used to neutralise an alkaline solution?

- A.** Nitrogen
- B.** Magnesium
- C.** Iron
- D.** Helium

(1 mark)

40 Poly(ethene) is an addition polymer.

Which term describes the monomer from which it is formed?

- A.** A saturated compound called ethane
- B.** A saturated compound called ethene
- C.** An unsaturated compound called ethane
- D.** An unsaturated compound called ethene

(1 mark)